

4MA1_2F_ch4_geometry_trigonometry

Nguyen Khac Kiem PhD

1. 4MA1/2FR_Summer_2024_Q26

The diagram shows a hexagon $ABCDEF$

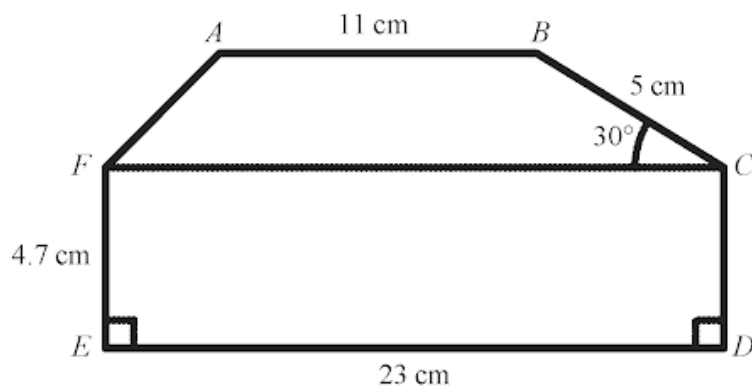


Diagram **NOT**
accurately drawn

Angle $BCF = 30^\circ$

AB , FC and ED are parallel.

Calculate the area of $ABCDEF$

Show your working clearly.

..... cm^2

(5)

2. 4MA1/2FR_Summer_2024_Q21

The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square $ABCD$

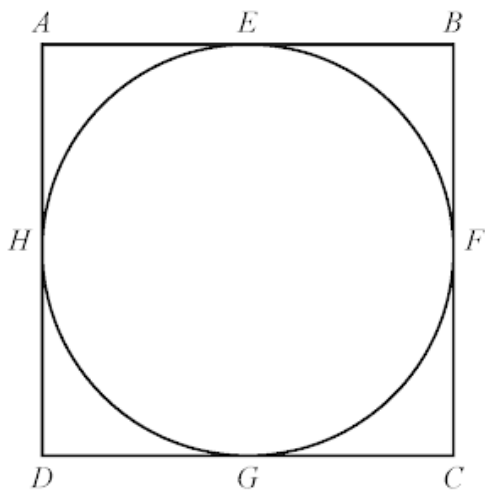


Diagram **NOT**
accurately drawn

The circle touches the square at the points E, F, G and H

The area of the square is 81 cm^2

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

..... cm
(4)

3. 4MA1/2FR_Summer_2024_Q12

The size of each exterior angle of a regular polygon is 15°
Work out the number of sides of the regular polygon.

(2)

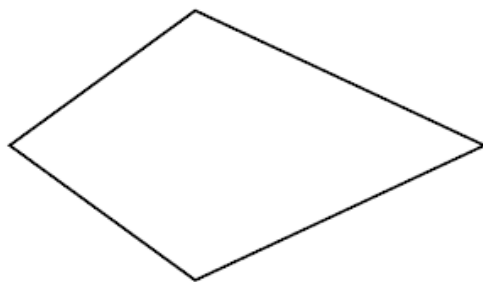
4. 4MA1/2FR_Summer_2024_Q10

Use a ruler and compasses only to construct a square with sides of length 7 cm.
You must show all your construction lines.
Two sides of the square have been drawn for you.



(3)

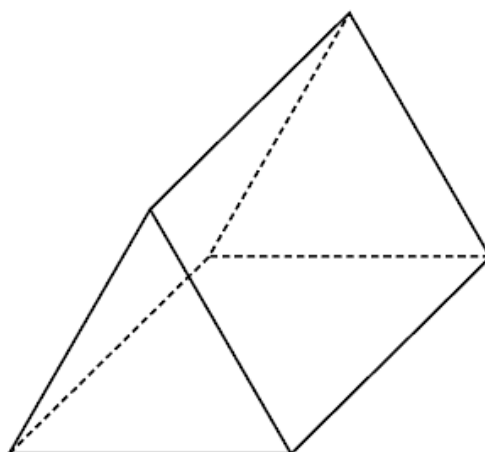
Here is a quadrilateral.



(a) What is the mathematical name for this type of quadrilateral?

(1)

The diagram shows a 3-D shape.



(b) (i) What is the mathematical name for this 3-D shape?

(1)

(ii) How many edges has this shape?

(1)

6. 4MA1/2F_Summer_2024_Q23

The diagram shows a 6-sided shape $ABCDEF$

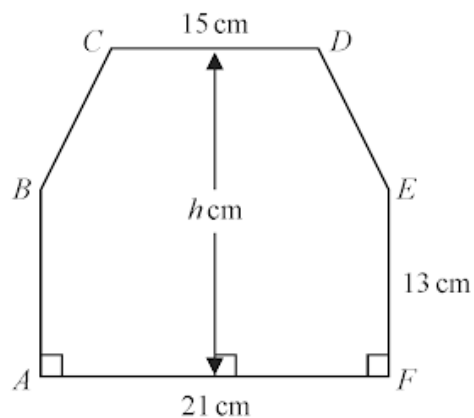


Diagram **NOT**
accurately drawn

$$AF = 21 \text{ cm} \quad CD = 15 \text{ cm} \quad AB = FE = 13 \text{ cm}$$

The perpendicular height of the shape is h cm
 CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

(4)

7. 4MA1/2F_Summer_2024_Q21

The diagram shows triangle ABC

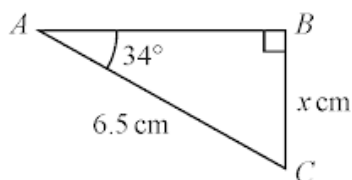
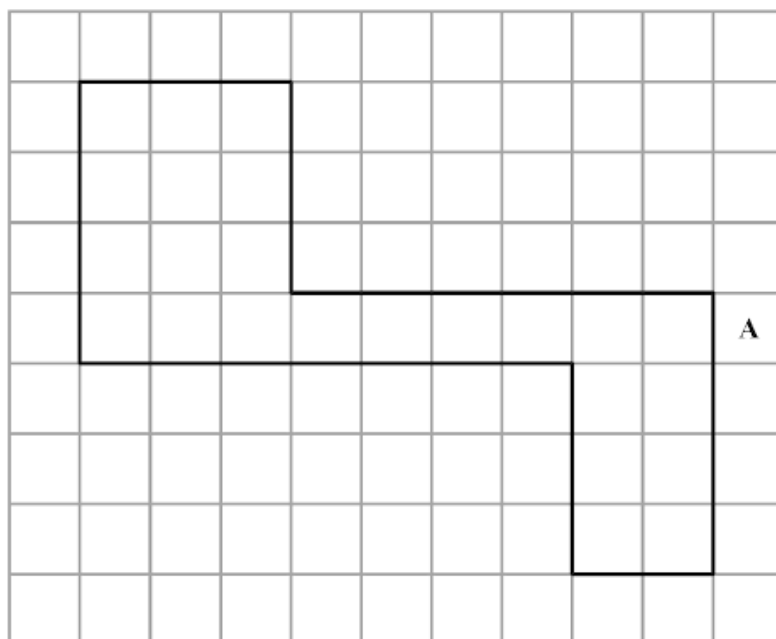


Diagram **NOT**
accurately drawn

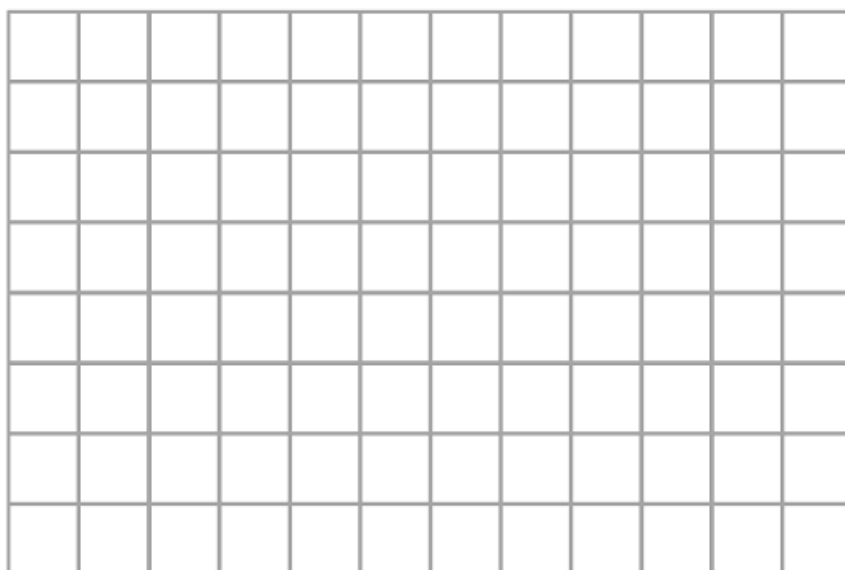
Work out the value of x
 Give your answer correct to one decimal place.

(3)

The diagram shows shape **A** drawn on a centimetre grid.



On the centimetre grid below, draw a rectangle that has the same area as shape **A**



(2)

9. 4MA1/2F_Summer_2024_Q9

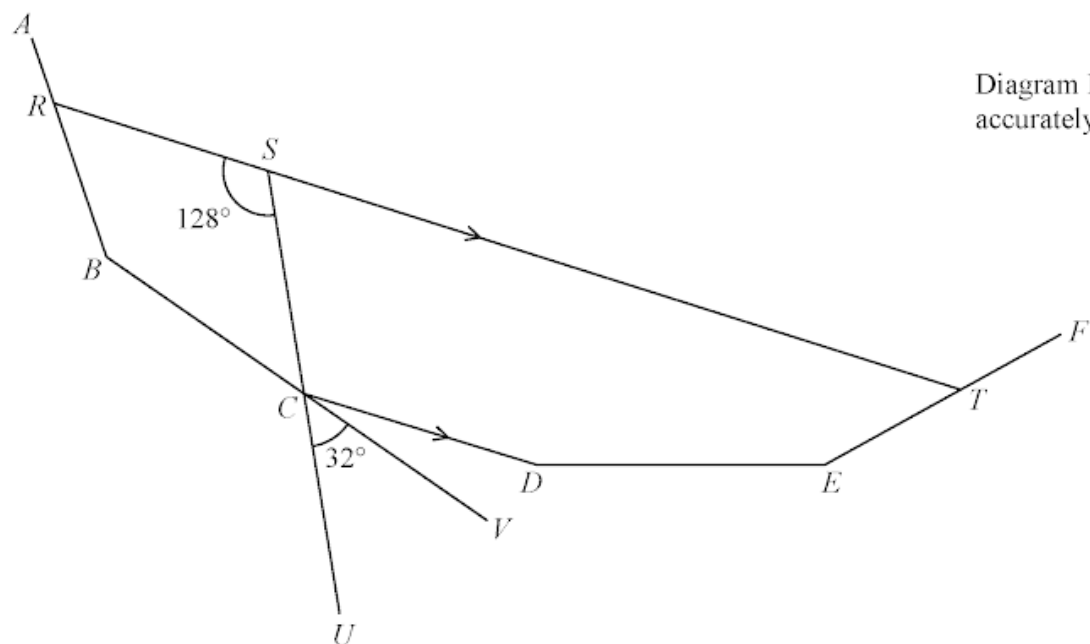
A circle has a radius of 9 cm.

Calculate the area of the circle.

Give your answer correct to 3 significant figures.

..... cm²

(2)



AB , BC , CD , DE and EF are five sides of a regular polygon.

RST , SCU and BCV are straight lines.

RST is parallel to CD

Angle $RSC = 128^\circ$

Angle $UCV = 32^\circ$

Work out how many sides the polygon has.

Show your working clearly.

4 marks

The diagram shows two solids, **A** and **B**, made from two different metals.

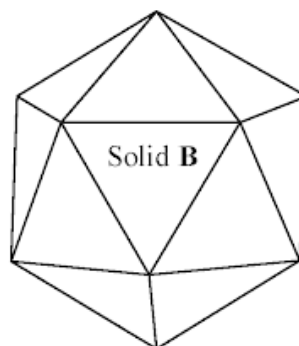
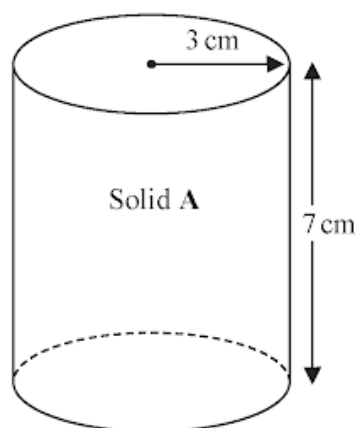


Diagram **NOT**
accurately drawn

Solid **A** is in the shape of a cylinder with radius 3 cm and height 7 cm

Solid **A** has a mass of 2000 g

Solid **B** has a mass of 3375 g

Solid **B** has a volume of 450 cm^3

All of the metal from Solid **A** and Solid **B** is melted down to make a uniform Solid **C**

Given that there is no change to mass or volume during this process

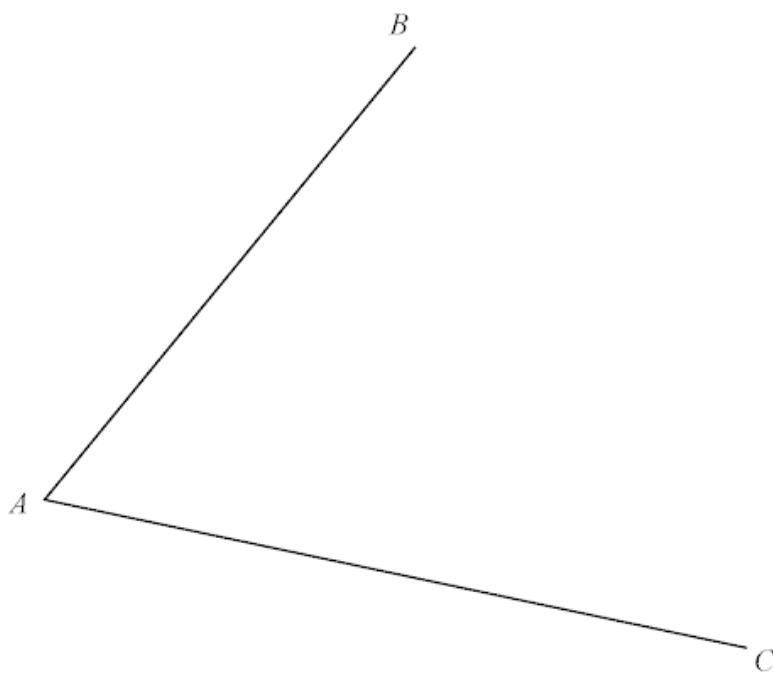
work out the density of Solid **C**

Give your answer correct to one decimal place.

..... g/cm^3

3 marks

Using ruler and compasses only, construct the bisector of angle BAC
You must show all your construction lines.



2 marks

The diagram shows the plan of a floor.

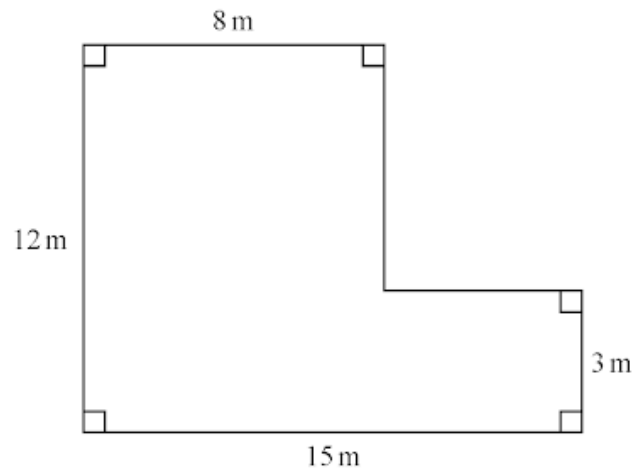


Diagram **NOT**
accurately drawn

Indira is going to paint the floor.

She needs to buy enough tins of paint to cover the floor with one coat of paint.

Each tin of paint covers an area of 7 m^2

Each tin of paint costs £23.90

Indira buys the least possible number of tins of paint.

Work out the total cost of the tins of paint that Indira buys.

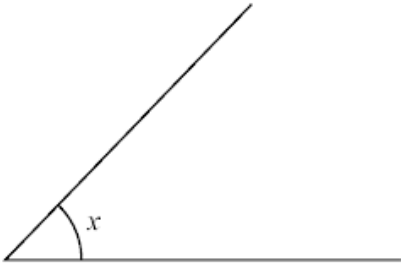
Show your working clearly.

£.....

5 marks

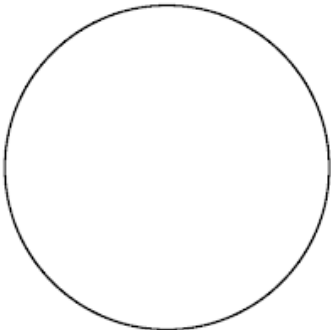
(a) Write down the mathematical name for an 8-sided polygon.

.....
(1)



(b) What type of angle is the angle marked x ?

.....
(1)



(c) On the diagram above, draw a chord of the circle.

(1)

(d) Change 3.6 metres into centimetres.

..... cm
(1)

.....

.....

.....

.....

.....

The diagram shows an isosceles triangle, with base length 24 cm.

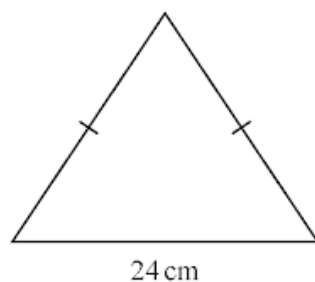


Diagram **NOT**
accurately drawn

The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

..... cm²

5 marks

The diagram shows two parallel lines AB and DEF

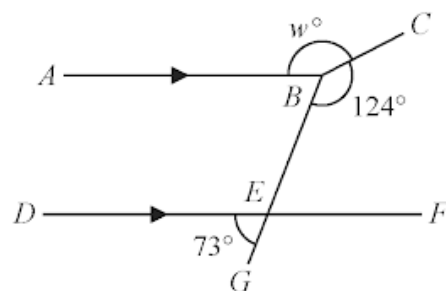


Diagram **NOT**
accurately drawn

BEG is a straight line.

$$\text{angle } DEG = 73^\circ \quad \text{angle } EBC = 124^\circ \quad \text{angle } ABC = w^\circ$$

Work out the value of w

Give reasons for each stage of your working.

$$w = \dots\dots\dots$$

4 marks

The diagram shows a solid prism.

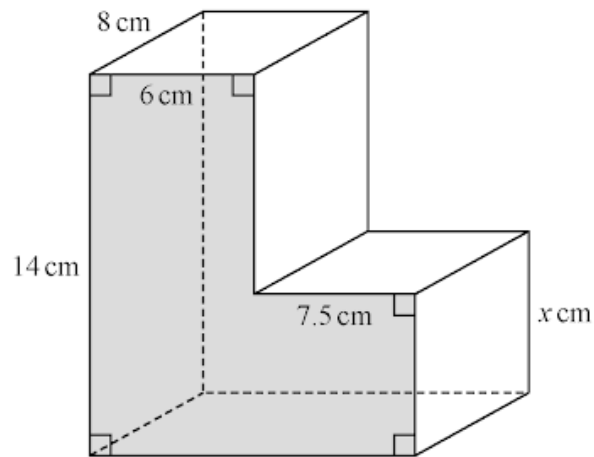


Diagram **NOT**
accurately drawn

The cross section of the prism is shown shaded.

The volume of the prism is 924 cm^3

Work out the value of x

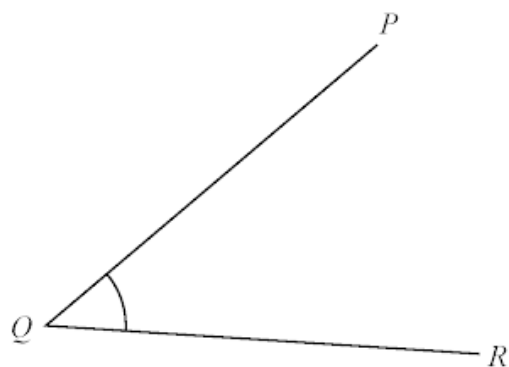
$x = \dots\dots\dots$

4 marks

(a) In the space below, draw a line of length 6.5 cm

(1)

The diagram shows the straight lines QP and QR



(b) Measure the size of angle PQR

°

(1)

The diagram shows a rectangular sheet of metal $ABCD$

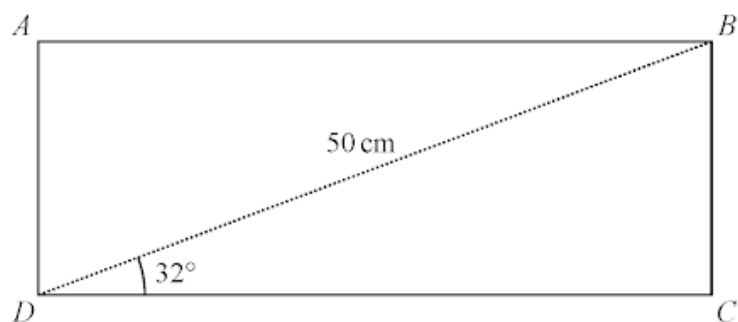


Diagram **NOT**
accurately drawn

$BD = 50$ cm and angle $BDC = 32^\circ$

Nasser joins side AD to side BC to form a cylinder.

BC is the height of the cylinder.

DC is the circumference of the cross section of the cylinder.

Work out the volume, in cm^3 , of the cylinder.

Give your answer correct to 3 significant figures.

A field is in the shape of a trapezium.

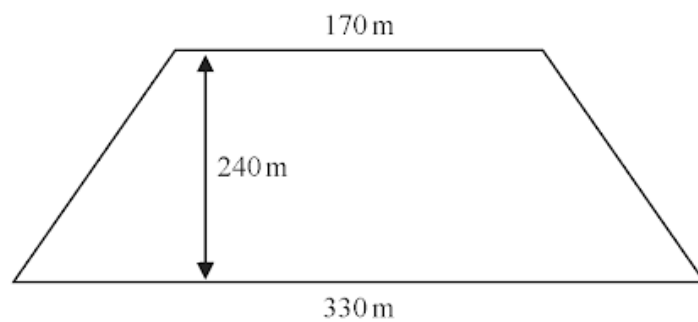


Diagram **NOT**
accurately drawn

The field is sold for a price of \$49 650

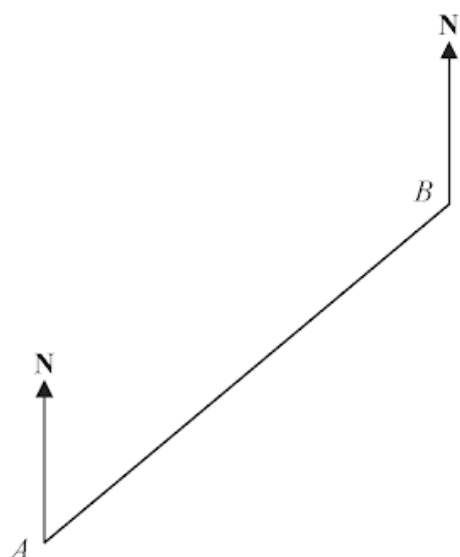
Given that 1 hectare = 10 000 m²

work out the average price of the field per hectare.

\$

4 marks

The scale drawing shows the positions of two boats, A and B



Scale: 1 cm represents 3 km

(a) Work out the actual distance, in kilometres, between A and B

..... km
(2)

Boat C is on a bearing of 110° from B

Boat C is 24 km from B

(b) On the scale drawing, mark with a cross ($\times$) the position of boat C

(3)

The diagram shows quadrilateral $ABCD$

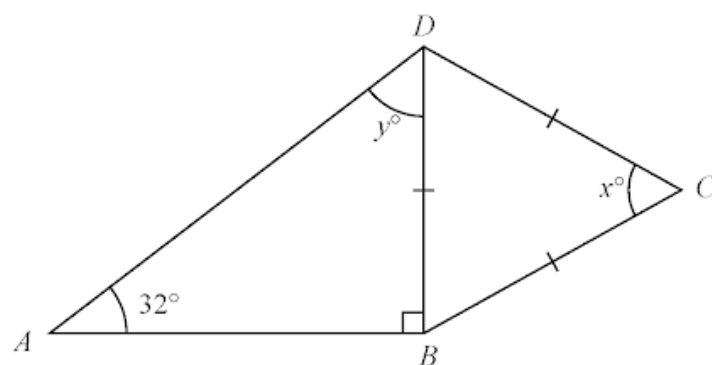


Diagram **NOT**
accurately drawn

$$BC = CD = DB$$

angle $DBA = 90^\circ$ and angle $DAB = 32^\circ$

(a) Work out the value of x

$$x = \dots\dots\dots$$

(1)

(b) (i) Work out the value of y

$$y = \dots\dots\dots$$

(1)

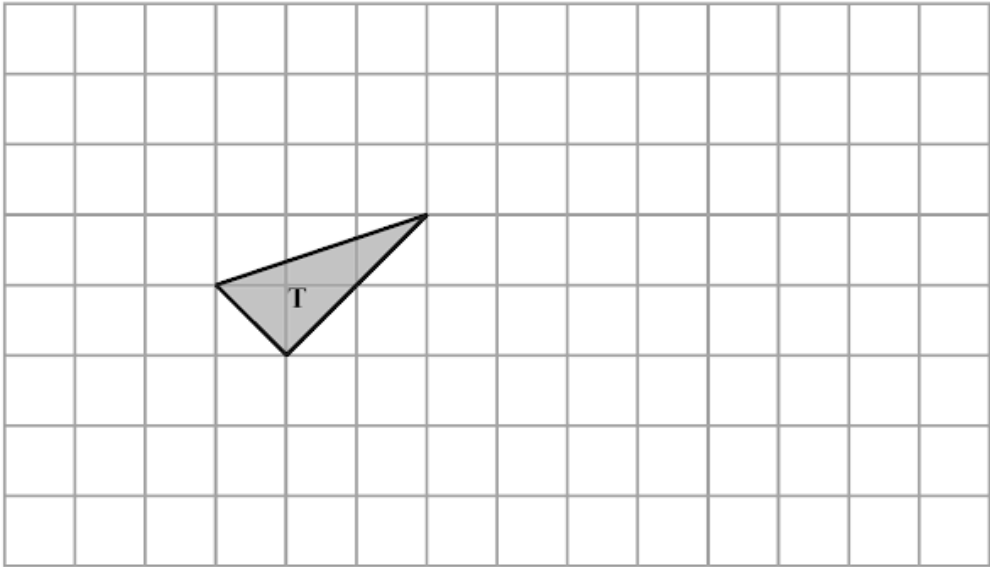
(ii) Give a reason for your answer to (b)(i).

.....

.....

(1)

(b) On the grid below, draw a shape that is congruent to shape **T**



(1)

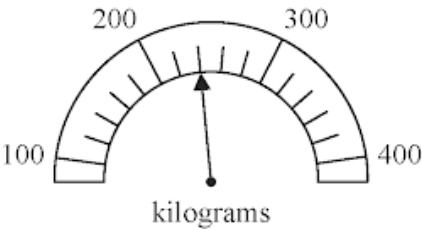
Here is a parallelogram.



(c) Write down the order of rotational symmetry of the parallelogram.

.....
(1)

Here is a scale.



(d) Write down the weight marked by the arrow.

..... kilograms
(1)

The diagram shows right-angled triangle ABD

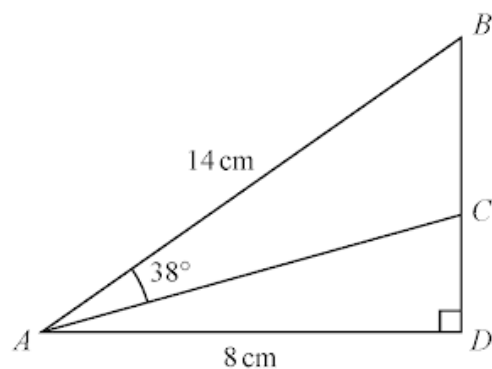


Diagram **NOT**
accurately drawn

$$AB = 14\text{ cm} \quad AD = 8\text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

4 marks

Here is a 9-sided regular polygon $ABCDEFGHIJ$, with centre O

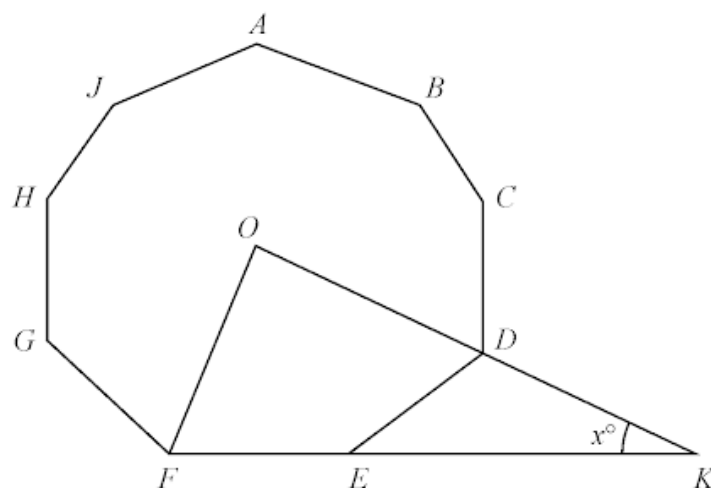


Diagram **NOT**
accurately drawn

ODK and FEK are straight lines.

Work out the value of x

$x =$

3 marks

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27. 4MA1/2F_Summer_2023_Q19

28. 4MA1/2F_Summer_2023_Q16

29. 4MA1/2F_Summer_2023_Q12

30. 4MA1/2F_Summer_2023_Q9

31. 4MA1/2F_Summer_2023_Q6

32. 4MA1/2FR_Winter_2022_Q26

33. 4MA1/2FR_Winter_2022_Q21

34. 4MA1/2FR_Winter_2022_Q17

35. 4MA1/2FR_Winter_2022_Q13

36. 4MA1/2FR_Winter_2022_Q9

37. 4MA1/2FR_Winter_2022_Q6

38. 4MA1/2FR_Winter_2022_Q3

39. 4MA1/2F_Winter_2022_Q26

40. 4MA1/2F_Winter_2022_Q20

41. 4MA1/2F_Winter_2022_Q19

42. 4MA1/2F_Winter_2022_Q18

43. 4MA1/2F_Winter_2022_Q12

44. 4MA1/2F_Winter_2022_Q9

45. 4MA1/2FR_Summer_2022_Q27

46. 4MA1/2FR_Summer_2022_Q22

47. 4MA1/2FR_Summer_2022_Q13

48. 4MA1/2FR_Summer_2022_Q11

49. 4MA1/2FR_Summer_2022_Q4

50. 4MA1/2F_Summer_2022_Q22

51. 4MA1/2F_Summer_2022_Q21

52. 4MA1/2F_Summer_2022_Q20

53. 4MA1/2F_Summer_2022_Q10

54. 4MA1/2F_Summer_2022_Q7

55. 4MA1/2F_Summer_2022_Q2

56. 4MA1/2FR_Winter_2021_Q12

57. 4MA1/2FR_Winter_2021_Q9

58. 4MA1/2FR_Winter_2021_Q4

59. 4MA1/2F_Winter_2021_Q22

60. 4MA1/2F_Winter_2021_Q14

61. 4MA1/2F_Winter_2021_Q13

62. 4MA1/2F_Winter_2021_Q11

63. 4MA1/2F_Winter_2021_Q10

64. 4MA1/2F_Winter_2021_Q6

65. 4MA1/2F_Summer_2021_Q24

66. 4MA1/2F_Summer_2021_Q21

67. 4MA1/2F_Summer_2021_Q19

68. 4MA1/2F_Summer_2021_Q17

69. 4MA1/2F_Summer_2021_Q11

70. 4MA1/2F_Summer_2021_Q10

71. 4MA1/2F_Summer_2021_Q2

72. 4MA1/2FR_Winter_2020_Q26

73. 4MA1/2FR_Winter_2020_Q19

74. 4MA1/2FR_Winter_2020_Q15

75. 4MA1/2FR_Winter_2020_Q5

76. 4MA1/2FR_Winter_2020_Q3

77. 4MA1/2F_Winter_2020_Q27

78. 4MA1/2F_Winter_2020_Q21

79. 4MA1/2F_Winter_2020_Q15

80. 4MA1/2F_Winter_2020_Q6

81. 4MA1/2F_Winter_2020_Q3

82. 4MA1/2FR_Summer_2020_Q26

83. 4MA1/2FR_Summer_2020_Q20

84. 4MA1/2FR_Summer_2020_Q7

85. 4MA1/2FR_Summer_2020_Q5

86. 4MA1/2FR_Summer_2020_Q2

87. 4MA1/2F_Summer_2020_Q24

88. 4MA1/2F_Summer_2020_Q23

89. 4MA1/2F_Summer_2020_Q22

90. 4MA1/2F_Summer_2020_Q18

91. 4MA1/2F_Summer_2020_Q13

92. 4MA1/2FR_Winter_2019_Q23

93. 4MA1/2FR_Winter_2019_Q22

94. 4MA1/2FR_Winter_2019_Q16

95. 4MA1/2FR_Winter_2019_Q14

96. 4MA1/2FR_Winter_2019_Q10

97. 4MA1/2F_Winter_2019_Q19

98. 4MA1/2F_Winter_2019_Q15

99. 4MA1/2F_Winter_2019_Q13

100. 4MA1/2F_Winter_2019_Q12

101. 4MA1/2F_Winter_2019_Q9

102. 4MA1/2FR_Summer_2019_Q27

103. 4MA1/2FR_Summer_2019_Q20

104. 4MA1/2FR_Summer_2019_Q16

105. 4MA1/2FR_Summer_2019_Q11

106. 4MA1/2FR_Summer_2019_Q10

107. 4MA1/2FR_Summer_2019_Q3

108. 4MA1/2F_Summer_2019_Q21

109. 4MA1/2F_Summer_2019_Q17

110. 4MA1/2F_Summer_2019_Q15

111. 4MA1/2F_Summer_2019_Q10

112. 4MA1/2F_Summer_2019_Q4

113. 4MA1/2F_Summer_2019_Q3

114. 4MA1/2FR_Summer_2018_Q25

115. 4MA1/2FR_Summer_2018_Q24

116. 4MA1/2FR_Summer_2018_Q23

117. 4MA1/2FR_Summer_2018_Q7

118. 4MA1/2F_Summer_2018_Q21

119. 4MA1/2F_Summer_2018_Q19

120. 4MA1/2F_Summer_2018_Q15

121. 4MA1/2F_Summer_2018_Q9
